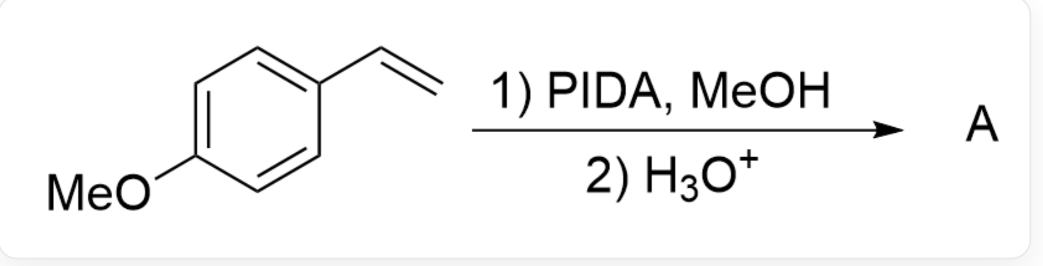
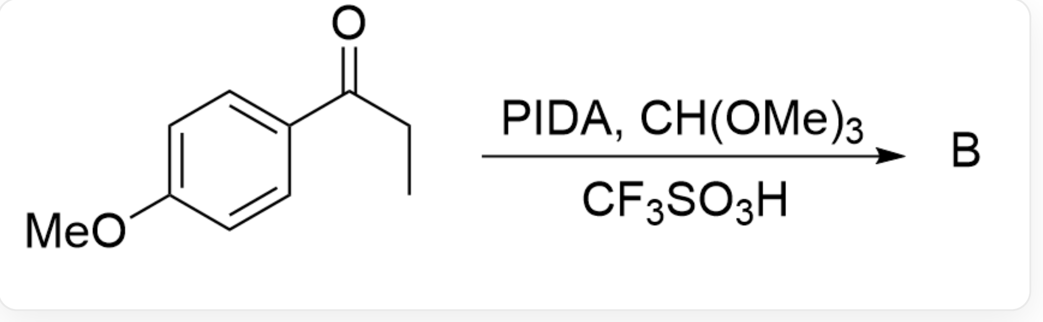
Question

PIDA is a commonly used relatively mild oxidizing agent.



C=CC1=CC=C(OC)C=C1>$PIDA.MeOH$>H_3O^+>[A],A is the reaction product

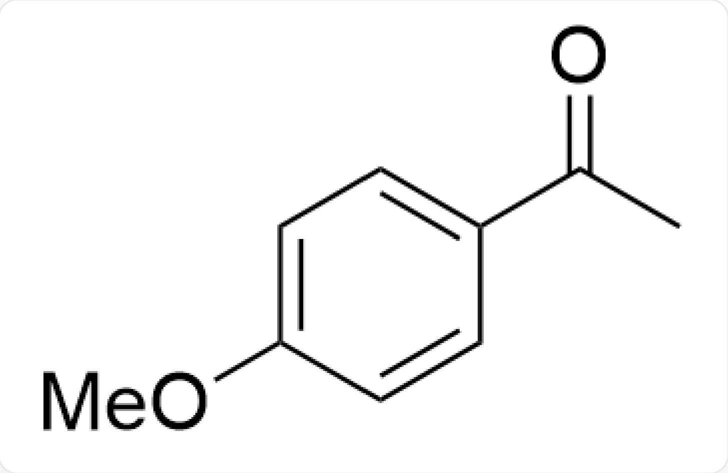


O=C(CC)C1=CC=C(OC)C=C1>$PIDA.CH(OMe)_3.CF_3SO_3H$>[B],B is the reaction product

Please try to predict the main products of the above two reactions.

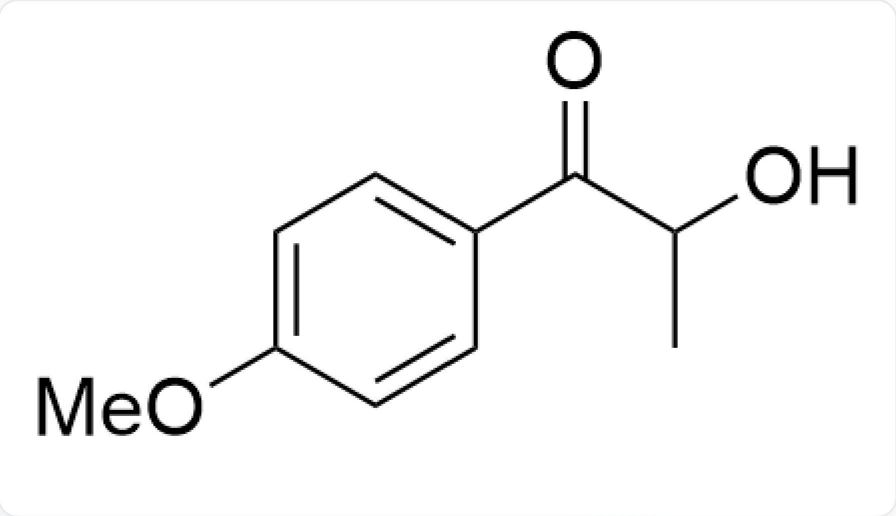
A. All other options are incorrect

B.



CC(C1=CC=C(OC)C=C1)=O,Product A

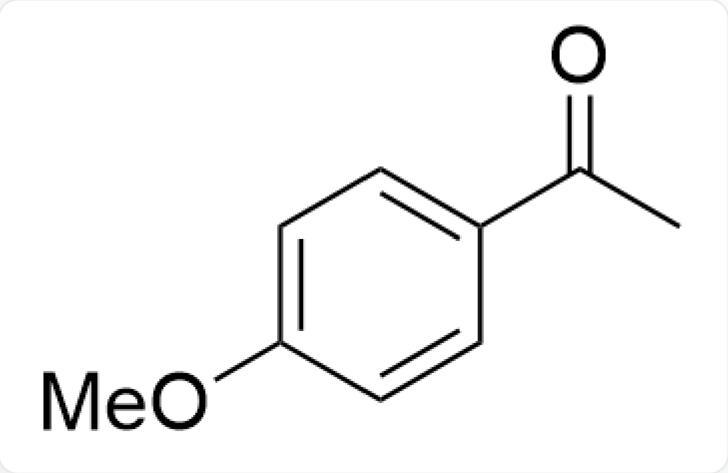
Product A



O=C(C(O)C)C1=CC=C(OC)C=C1,Product B

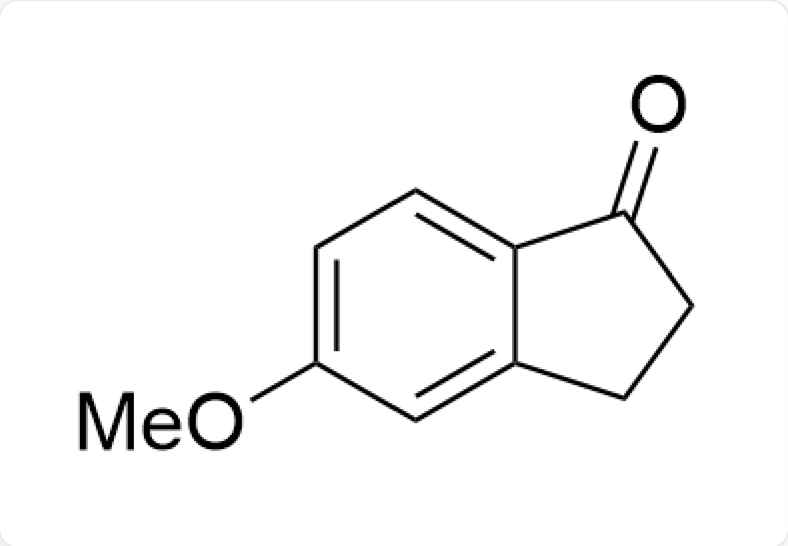
Product B

C.



CC(C1=CC=C(OC)C=C1)=O,Product A

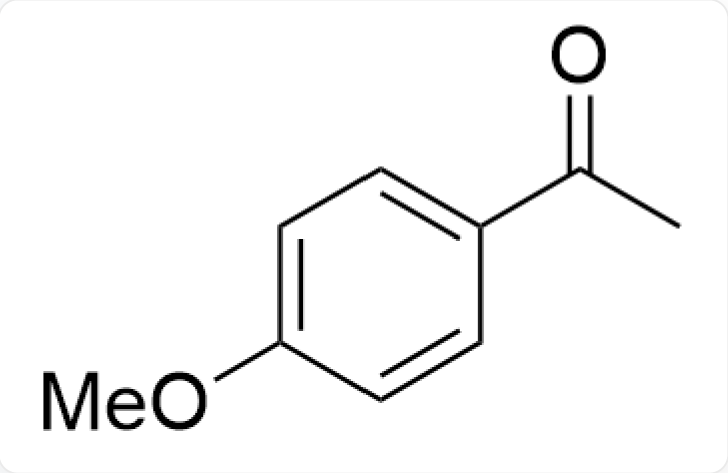
Product A



O=C1C2=CC=C(OC)C=C2CC1,Product B

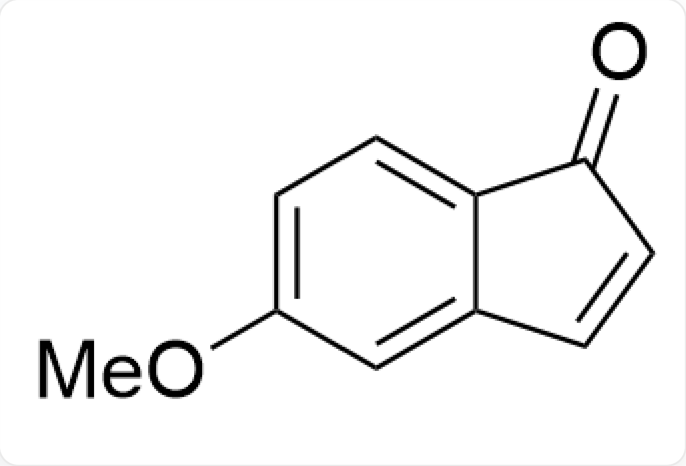
Product B

D.



CC(C1=CC=C(OC)C=C1)=O,Product A

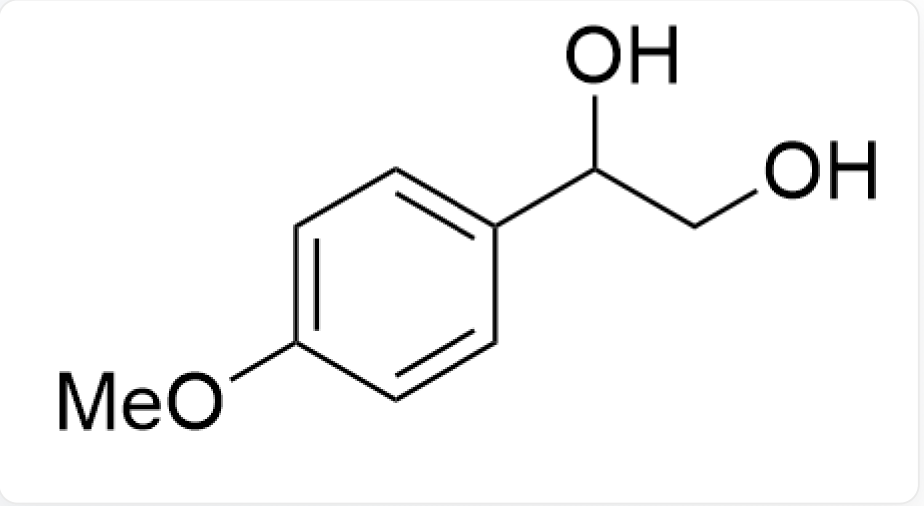
Product A



O=C1C2=CC=C(OC)C=C2C=C1,Product B

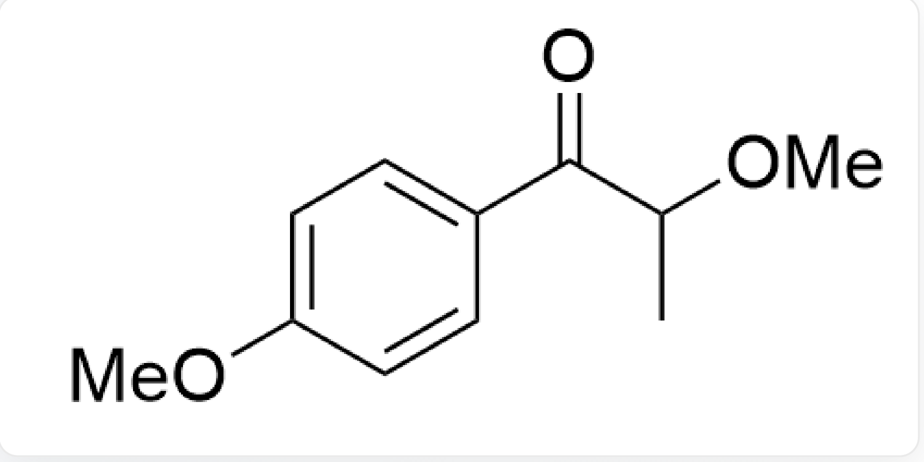
Product B

E.



OC(CO)C1=CC=C(OC)C=C1,Product A

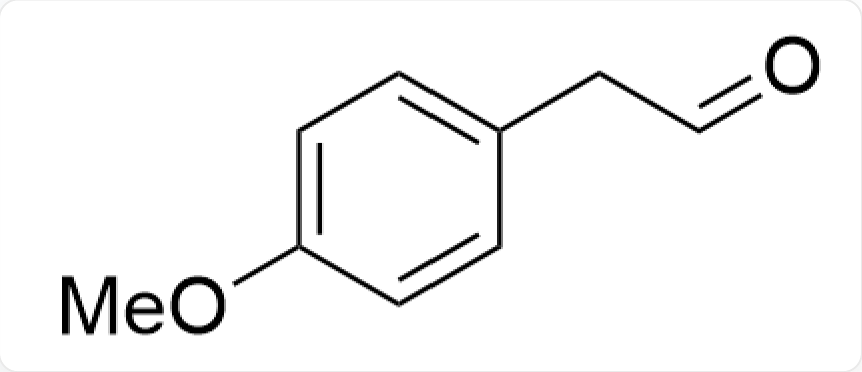
Product A



O=C(C(OC)C)C1=CC=C(OC)C=C1,Product B

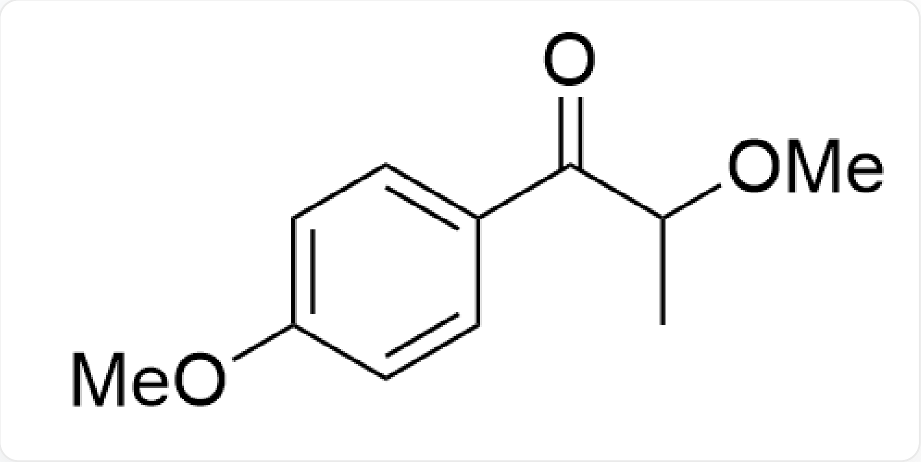
Product B

F.



O=CCC1=CC=C(OC)C=C1,Product A

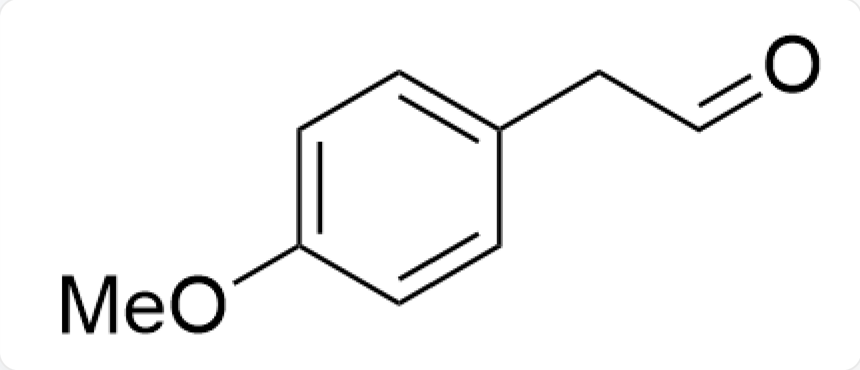
Product A



O=C(C(OC)C)C1=CC=C(OC)C=C1,Product B

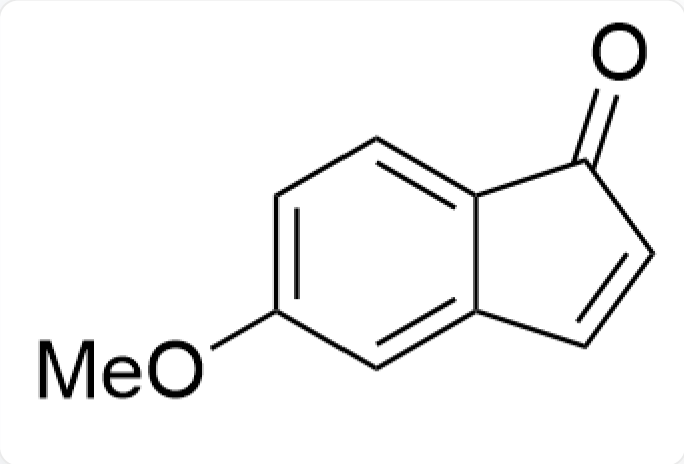
Product B

G.



O=CCC1=CC=C(OC)C=C1, Product A

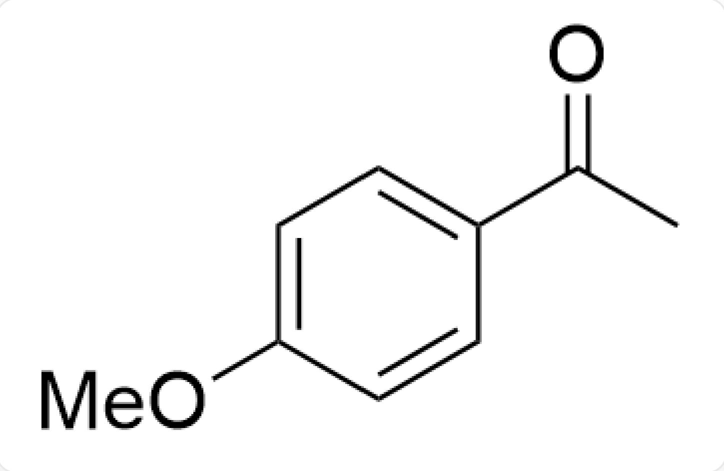
Product A



O=C1C2=CC=C(OC)C=C2C=C1, Product B

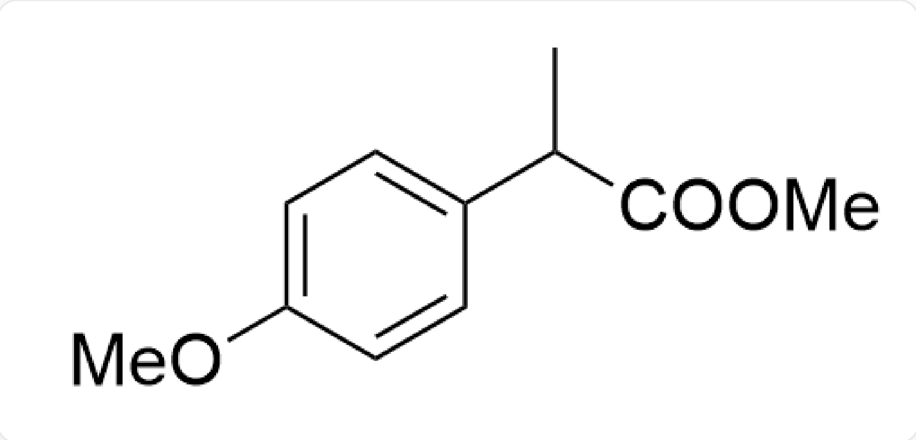
Product B

H.



CC(C1=CC=C(OC)C=C1)=O,Product A

Product A



CC(C(OC)=O)C1=CC=C(OC)C=C1,Product B

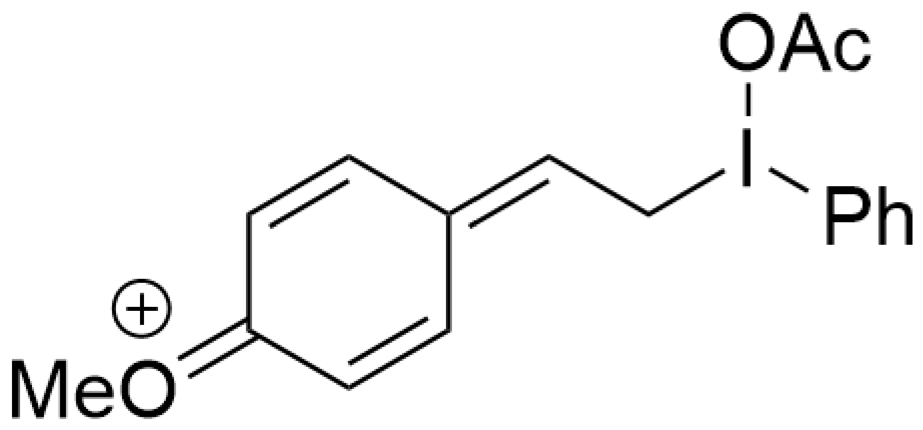
Product B

Answer

Correct Answer: A

Detailed Explanation

For Reaction 1, the substrate first reacts with PIDA to generate an intermediate

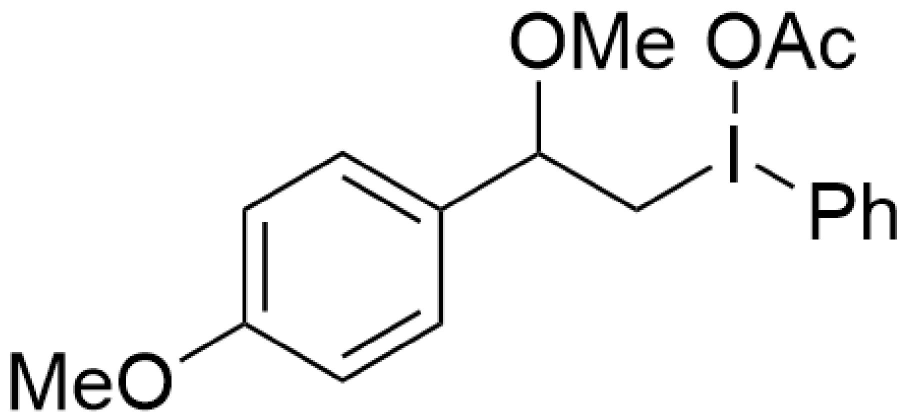


C[O+]=C(C=C/1)C=CC1=C\CI(C2=CC=CC=C2)OC(C)=O

CHECKPOINT 1 PTS

The substrate first reacts with PIDA to generate the intermediate C[O+]=C(C=C/1)C=CC1=C\CI(C2=CC=CC=C2)OC(C)=O

Subsequently, it may further react with methanol to form the intermediate



COC1=CC=C(C(OC)CI(C2=CC=CC=C2)OC(C)=O)C=C1

CHECKPOINT 1 PTS

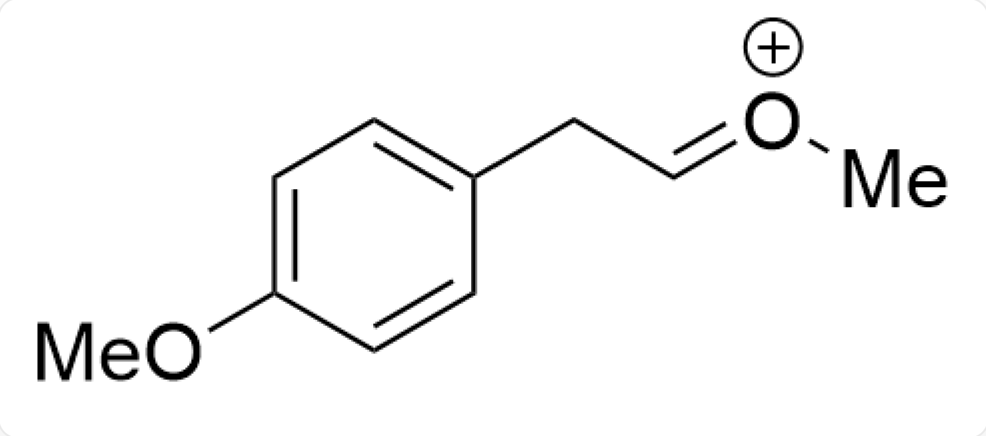
Further reaction with methanol forms the intermediate COC1=CC=C(C(OC)CI(C2=CC=CC=C2)OC(C)=O)C=C1

The electron-rich benzene ring preferentially migrates over hydrogen

CHECKPOINT 1 PTS

The electron-rich benzene ring preferentially migrates over hydrogen

to yield the intermediate

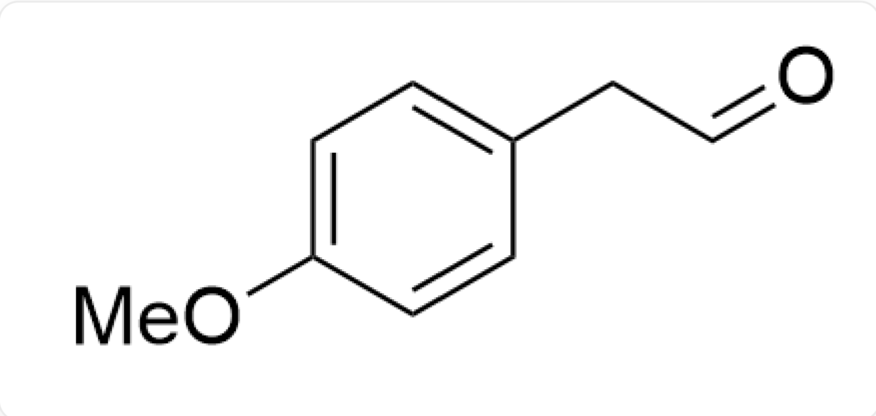


COC1=CC=C(C/C=[O+]/C)C=C1

CHECKPOINT 1 PTS

Benzene ring migration yields the intermediate COC1=CC=C(C/C=[O+]/C)C=C1

The resulting intermediate undergoes hydrolysis to give the final aldehyde product

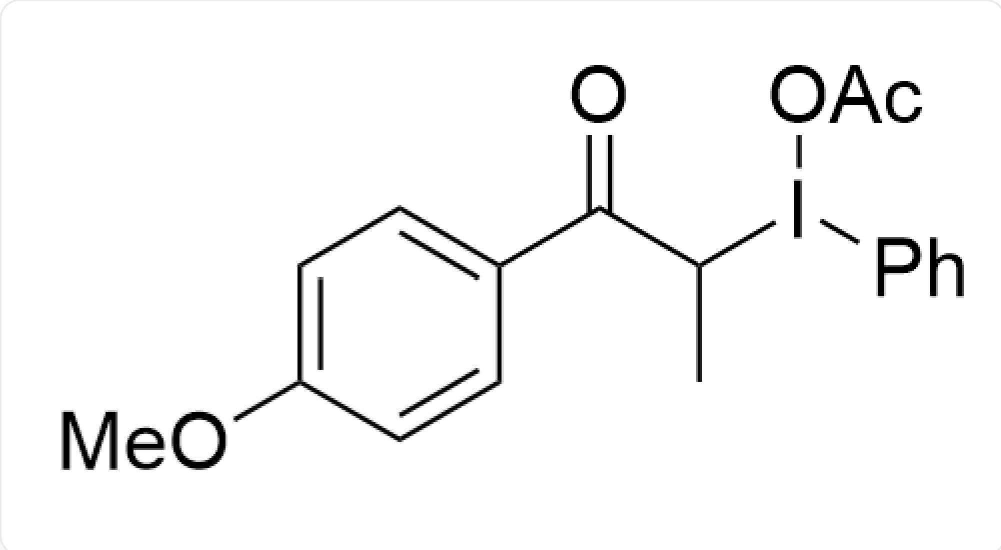


O=CCC1=CC=C(OC)C=C1

CHECKPOINT 2 PTS

The intermediate undergoes hydrolysis to yield product **A**, O=CCC1=CC=C(OC)C=C1

For Reaction 2, the first possible intermediate is



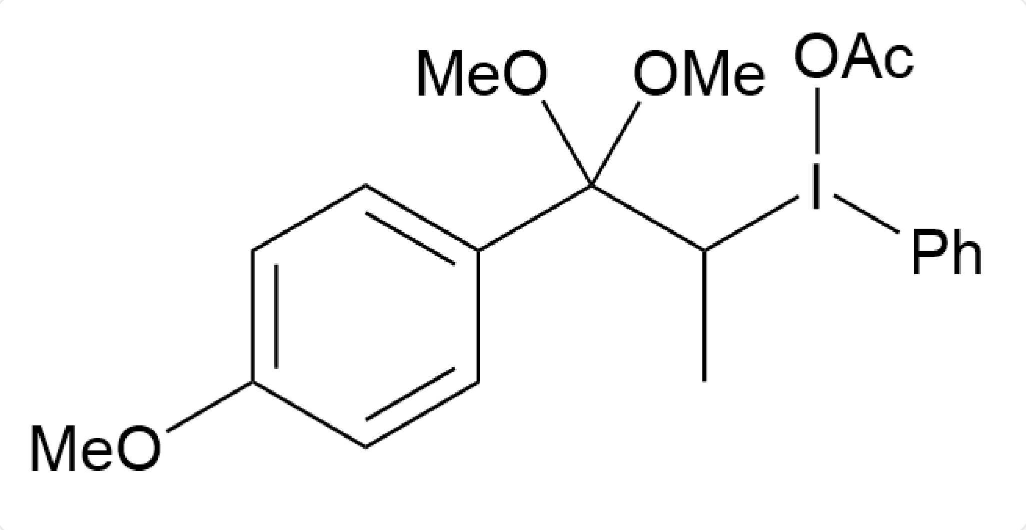
O=C(C(I(OC(C)=O)C1=CC=CC=C1)C)C2=CC=C(OC)C=C2

CHECKPOINT 1 PTS

The intermediate

O=C(C(I(OC(C)=O)C1=CC=CC=C1)C)C2=CC=C(OC)C=C2

Subsequently, it undergoes further esterification to give the intermediate



CC(I(OC(C)=O)C1=CC=CC=C1)C(OC)(OC)C2=CC=C(OC)C=C2

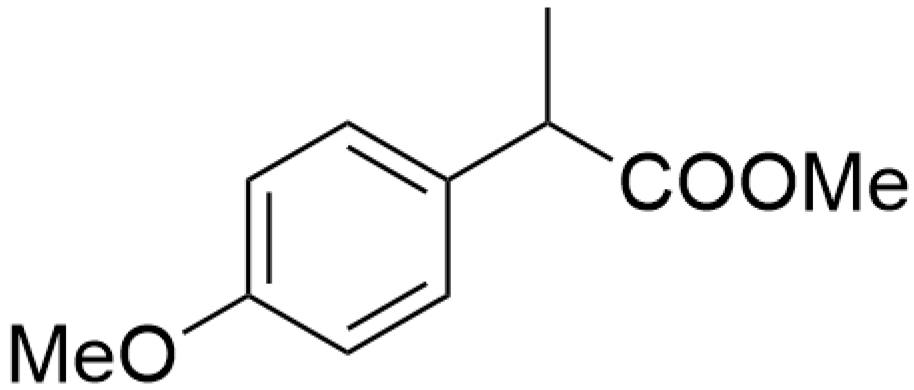
CHECKPOINT 1 PTS

Further esterification yields the intermediate CC(I(OC(C)=O)C1=CC=CC=C1)C(OC)(OC)C2=CC=C(OC)C=C2

Finally, the benzene ring migrates, and hydrolysis gives the final product

CHECKPOINT 1 PTS

Benzene ring migration and hydrolysis yield the final product **B**



CC(C(OC)=O)C1=CC=C(OC)C=C1, product B

CHECKPOINT 2 PTS

Product **B** is CC(C(OC)=O)C1=CC=C(OC)C=C1

Thus, none of the options match the structure, and option A is correct.